Introduction to Computing (CS 101)

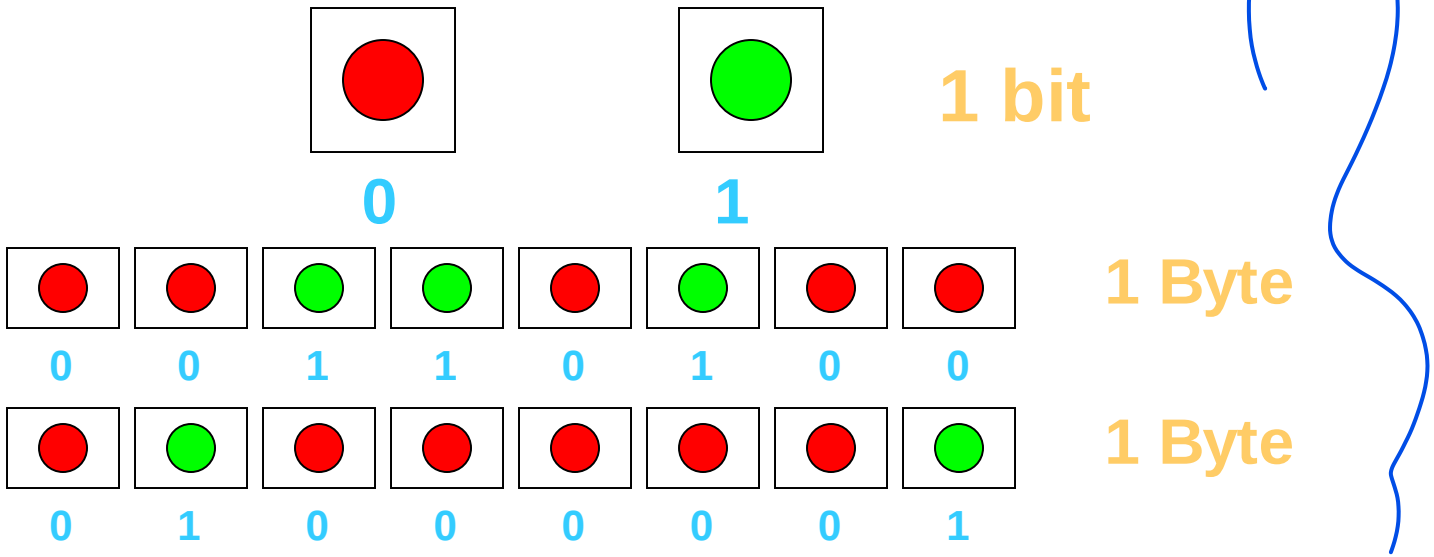
Hexadecimal Notation and Floating Point Number Representation

T. Venkatesh & John Jose

Department of Computer Science & Engineering

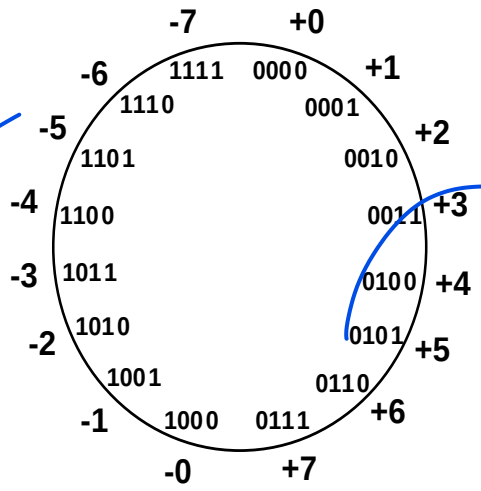
Indian Institute of Technology Guwahati

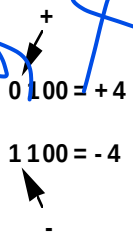
Review of Binary Storage



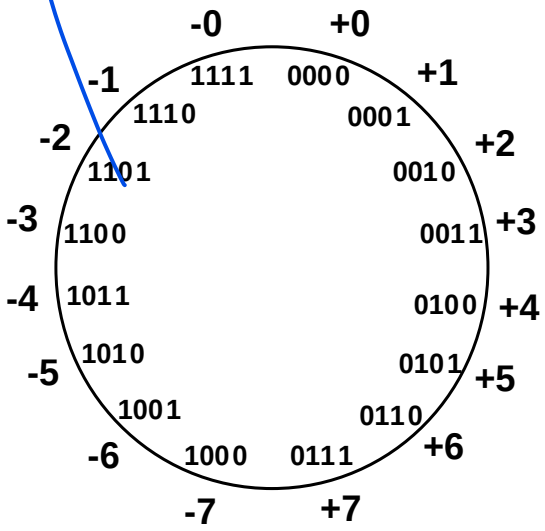
- Bits
- Bytes

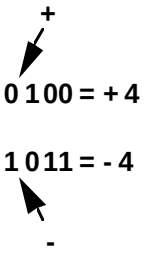
Review of Number Representation



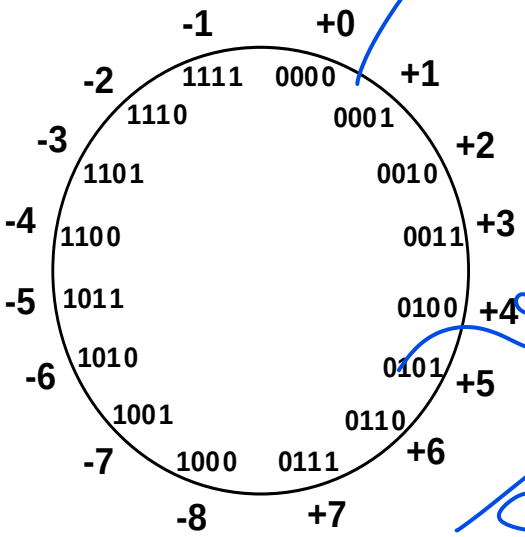


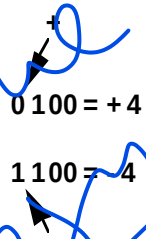
SM Form





1's C





2's C

Positive Integers

- Decimal system: made of 10 digits, {0,1,2, . . . , 9}

$$41 = 4\times10^1 + 1\times10^0$$

$$255 = 2\times10^2 + 5\times10^1 + 5\times10^0$$

- Binary system: made of two digits, {0,1}

$$00101001 = 0\times2^7 + 0\times2^6 + 1\times2^5 + 0\times2^4 + 1\times2^3 + 0\times2^2 + 0\times2^1 + 1\times2^0$$

$$= 32 + 8 + 1 = 41$$

- Hexadecimal System: the digits in base 16 are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, and F

  0x as a prefix is the convention for hexadecimal number

  Ex: 0x4B2A

Hexadecimal System

Denary/Decimal	Binary	Hexadecimal
Base 10 Number System	Base 2 Number System	Base 16 Number System
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

0x428 = ?

$= 4x16^2+2x16^1+ 8x16^0$

= 1024+32+8 = 1064

0xAD4 = ?

$= 10x16^2+13x16^1+ 4x16^0$

= 2560+208+4 = 2772

Converting Binary to Hexadecimal

- Mark groups of four (from right)
- Convert each group

10101011

10101011

A B

10101011 is AB in base 16 or **0xAB**

1010001101011110

1010001101011110

A 3 5 E

Hence, 1010001101011110 = 0xA35E

Denary/Decimal	Binary	Hexadecimal
Base 10 Number System	Base 2 Number System	Base 16 Number System
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Integers and Real Numbers

- **Integers: the universe is infinite but discrete**
  - No numbers between consecutive integers, e.g., 5 and 6
  - A countable (finite) number of items in a finite range
  - Referred to as fixed-point numbers
- **Real numbers – the universe is infinite and continuous**
  - Fractions represented by decimal notation
    - Rational numbers, e.g., 5/2 = 2.5
    - Irrational numbers, e.g., 22/7 = 3.14159265 . . .
  - Infinite numbers exist even in the smallest range
  - Referred to as floating-point numbers

Wide Range of Numbers

- A large number:

$$976{,}000{,}000{,}000{,}000 = 9.76 \times 10^{14}$$

- A small number:

$$0.000000000000976 = 9.76 \times 10^{-14}$$

Scientific Notation

- Decimal numbers
  - 0.513×10^5, 5.13×10^4 and 51.3×10^3 are written in scientific notation.
  - 5.13×10^4 is the normalized scientific notation.
- Binary numbers
  - Base 2
  - Binary point – multiplication by 2 moves the point to the right.
  - Normalized scientific notation, e.g., $1.0_2 \times 2^{-1}$

Floating Point Numbers

- General format

$$\pm 1.bbbbb_2 \times 2^{eeee}$$

or $$(-1)^S \times (1+F) \times 2^E$$

- Where
  - S = sign, 0 for positive, 1 for negative
  - F = fraction (or mantissa) as a binary integer, 1+F is called significand
  - E = exponent as a binary integer, positive or negative (two's complement ?? / SM ??)

Binary to Decimal Conversion

Binary $(-1)^S (1.b_1b_2b_3b_4) \times 2^E$

Decimal $(-1)^S \times (1 + b_1\times2^{-1} + b_2\times2^{-2} + b_3\times2^{-3} + b_4\times2^{-4}) \times 2^E$

Example: -1.1100×2^{-2} (binary)

$= - (1 + 2^{-1} + 2^{-2}) \times 2^{-2}$

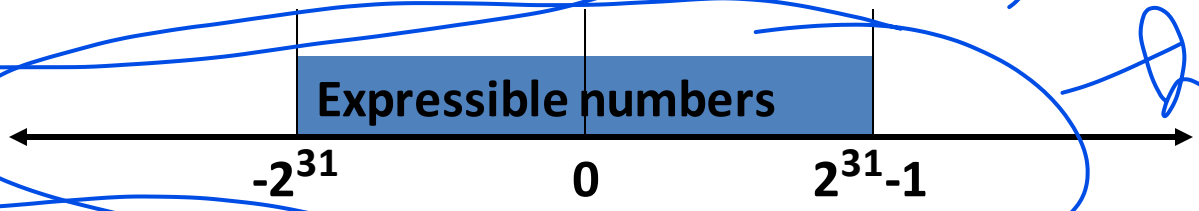
$= - (1 + 0.5 + 0.25)/4$

$= - 1.75/4$

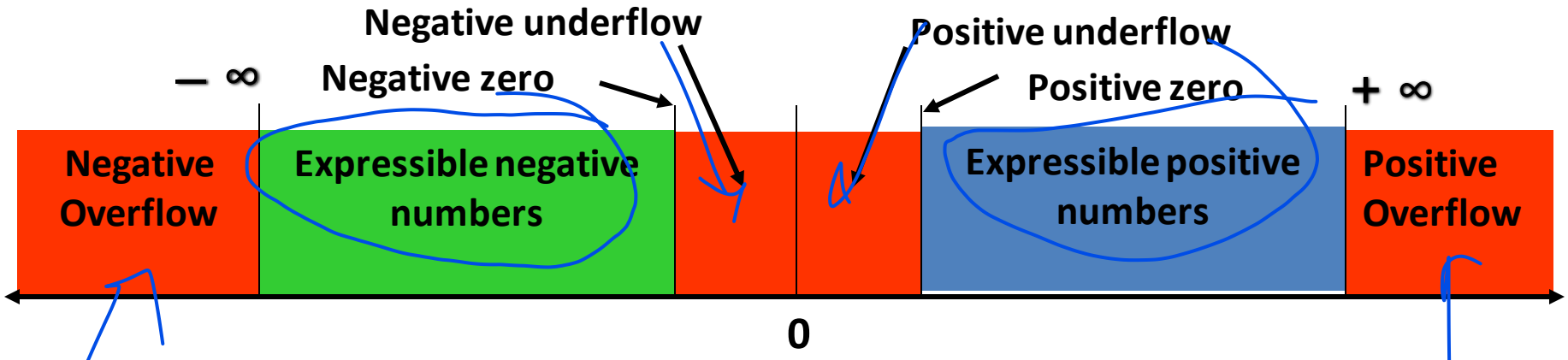
$= - 0.4375$ (decimal)

Numbers in 32-bit Formats

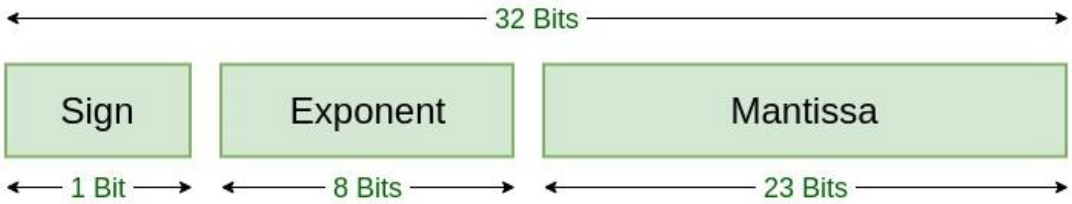
- Two's complement integers



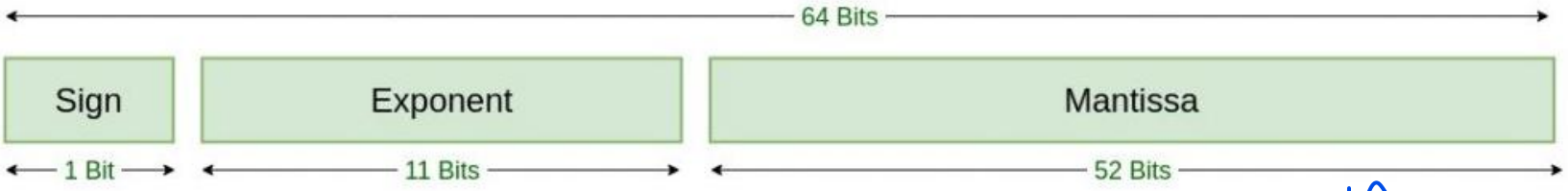
- Floating point numbers



IEEE 754 Floating Point Standards



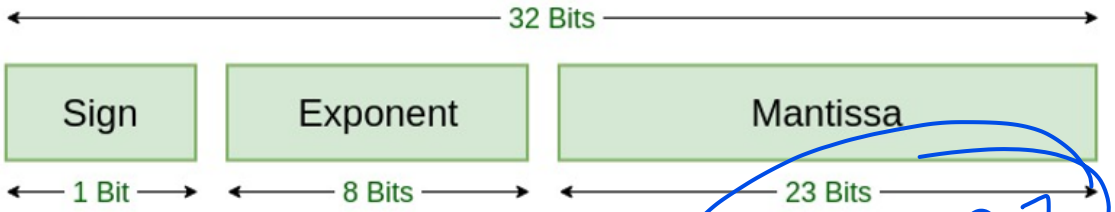
Single Precision
IEEE 754 Floating-Point Standard



Double Precision
IEEE 754 Floating-Point Standard

IEEE 754 Floating Point Standard

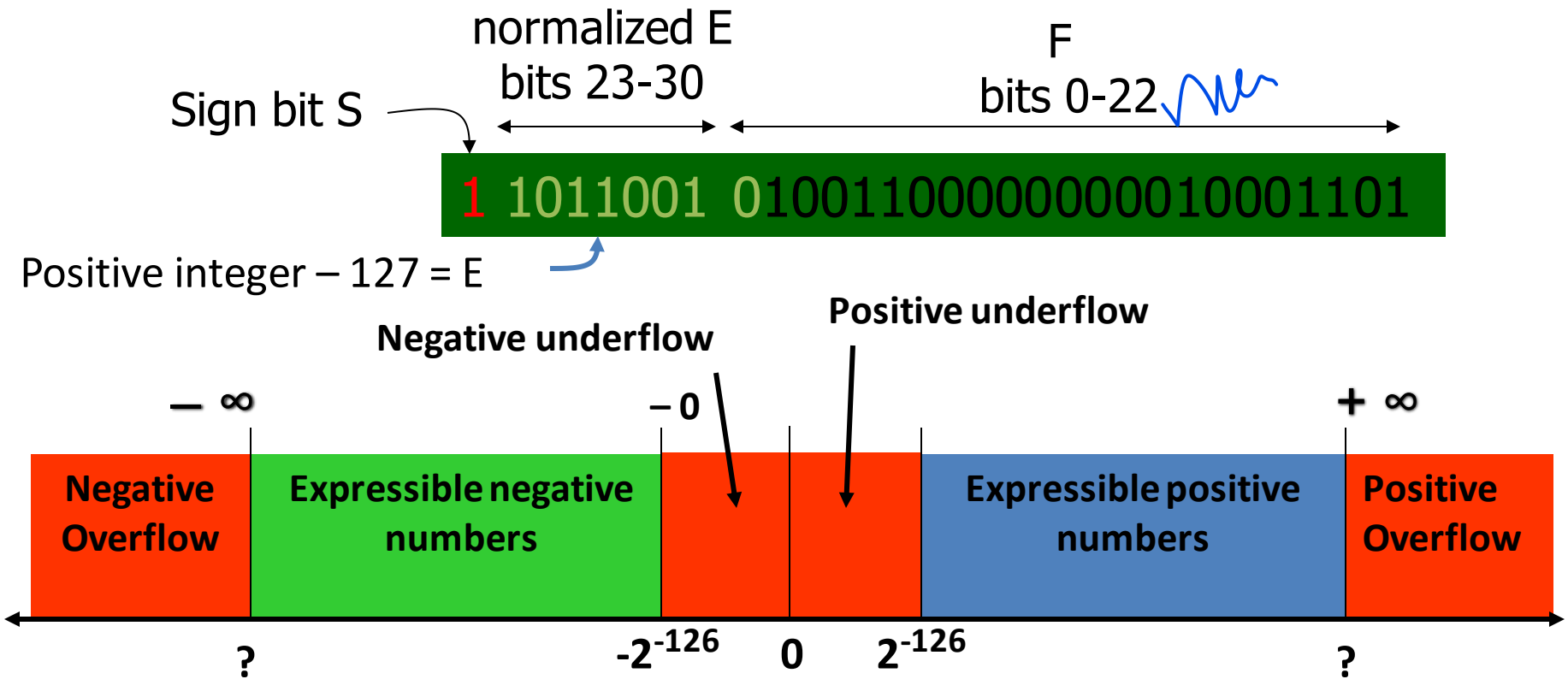
- Single Precision Floating point numbers



- Biased exponent: true exponent range
  - [-126,127] is changed to [1, 254]
  - Biased exponent is an 8-bit positive binary integer.
  - True exponent obtained by subtracting 127ten or 01111111two
- First bit of significand is always 1
  - ± 1.bbbb . . . b × 2E
  - 1 before the binary point is implicitly assumed.
  - Significand field represents 23 bit fraction after the binary point.

IEEE 754 Floating Point Standard

- Single Precision Floating point numbers



Examples

Biased exponent (1-254), bias 127 (01111111) to be subtracted

$1.1010001 \times 2^{10100} = 0\ 10010011\ 10100010000000000000000 = 1.6328125 \times 2^{20}$

$-1.1010001 \times 2^{10100} = 1\ 10010011\ 10100010000000000000000 = -1.6328125 \times 2^{20}$

$1.1010001 \times 2^{-10100} = 0\ 01101011\ 10100010000000000000000 = 1.6328125 \times 2^{-20}$

$-1.1010001 \times 2^{-10100} = 1\ 01101011\ 10100010000000000000000 = -1.6328125 \times 2^{-20}$

1.0
0.5
0.125
0.0078125
1.6328125

Sign bit

8-bit biased exponent
107 – 127
= – 20

23-bit Fraction (F)
of significand

Conversion From Hex to Decimal

- R1= 0x42200000

0 100 0010 0010 0000 0000 0000 0000 0000 = + 1.010 x 2^5

101000 = +40

- R2=0xC1200000

1 100 0001 0010 0000 0000 0000 0000 0000 = - 1.01 x 2^3

1010 = -10

- R3=0xC0800000

1 100 0000 1000 0000 0000 0000 0000 0000 = - 1.00 x 2^2

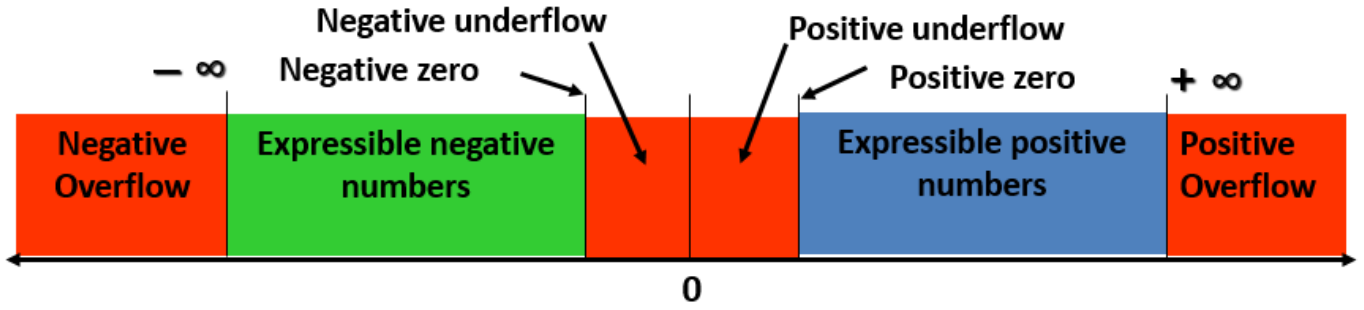
0100 = -4

Positive Zero in IEEE 754

0 00000000 00000000000000000000000

Biased exponent **Fraction**

- $+ 1.0 \times 2^{-127}$
- Smaller than the smallest positive number in single-precision IEEE 754 standard.
- Interpreted as positive zero.
- True exponent less than –126 is positive underflow; can be regarded as zero.



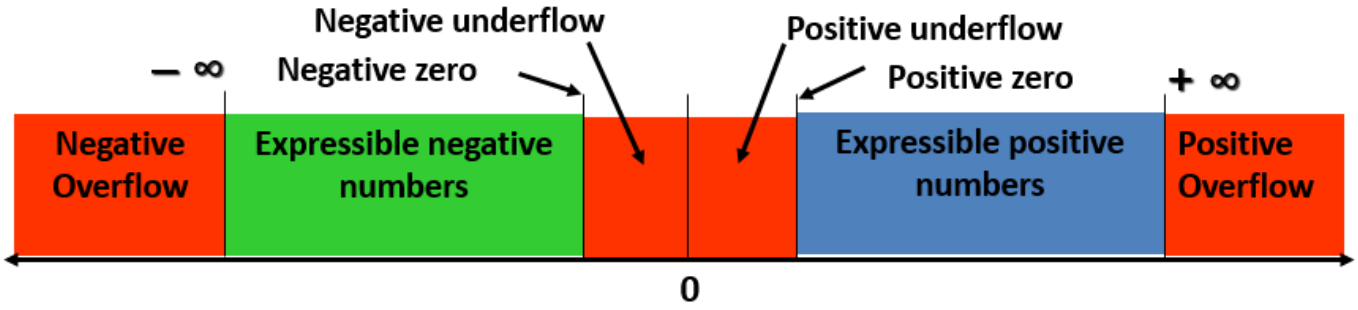
Negative Zero in IEEE 754

1 00000000 00000000000000000000000

Biased exponent

Fraction

- -1.0×2^{-127}
- Greater than the largest negative number in single-precision IEEE 754 standard.
- Interpreted as negative zero.
- True exponent less than –126 is negative underflow; may be regarded as 0.

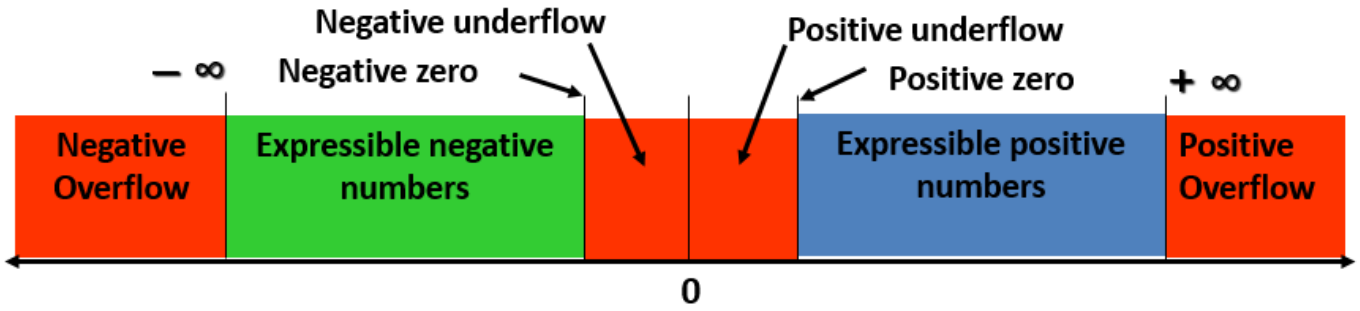


Positive Infinity in IEEE 754

0 11111111 00000000000000000000000

Biased exponent **Fraction**

- $+ 1.0 \times 2^{128}$
- Greater than the largest positive number in single-precision IEEE 754 standard.
- Interpreted as $+ \infty$
- If true exponent > 127, then the number is greater than ∞. It is called “not a number” or NaN and may be interpreted as ∞.

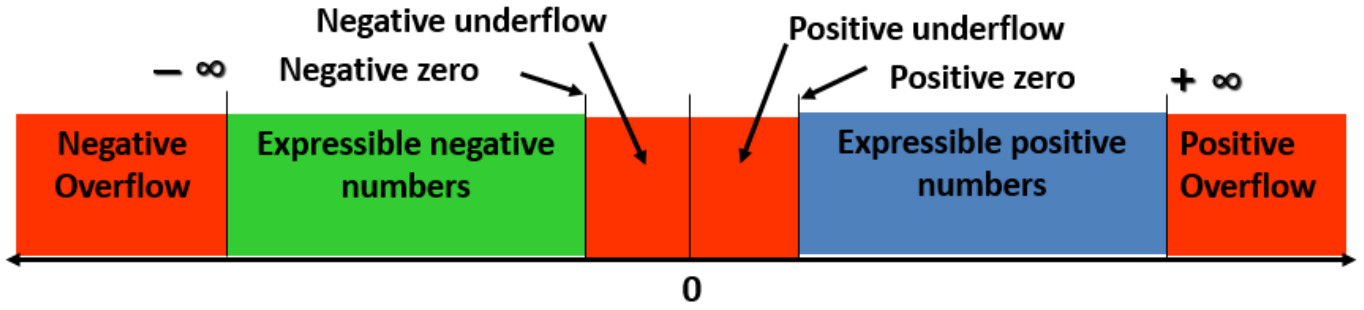


Negative Infinity in IEEE 754

1 11111111 00000000000000000000000

Biased exponent **Fraction**

- -1.0×2^{128}
- Smaller than the smallest negative number in single-precision IEEE 754 standard.
- Interpreted as $-\infty$
- If true exponent > 127, then the number is less than $-\infty$. It is called "not a number" or NaN and may be interpreted as $-\infty$.



FP Addition and Subtraction

1. **Significand alignment:** Right shift significand of smaller exponent until two exponents match.

2. **Addition:** Add significands and report error if overflow occurs. If significand = 0, return result as 0.

3. **Normalization**

    -Shift significand bits to normalize.

    - report overflow or underflow if exponent goes out of range.

4. **Rounding**

Example (4 Significant Fraction Bits)

- Subtraction: $0.5_{ten} - 0.4375_{ten}$
  - Floating point numbers to be added

    $1.000_{two} \times 2^{-1}$ and $-1.110_{two} \times 2^{-2}$
  - Significand of lesser exponent is shifted right until exponents match

    $-1.110_{two} \times 2^{-2} \rightarrow -0.111_{two} \times 2^{-1}$
  - Add significands, $1.000_{two} + (-0.111_{two})$

    Result is $0.001_{two} \times 2^{-1}$

  - Normalize, $1.000_{two} \times 2^{-4}$

    No overflow/underflow since

    $127 \geq \text{exponent} \geq -126$

    $1.000_{two} \times 2^{-4} = (1+0)/16 = 0.0625_{ten}$

```
 01000
+11001
 00001
```

2's complement addition, one bit added for sign

Thanks